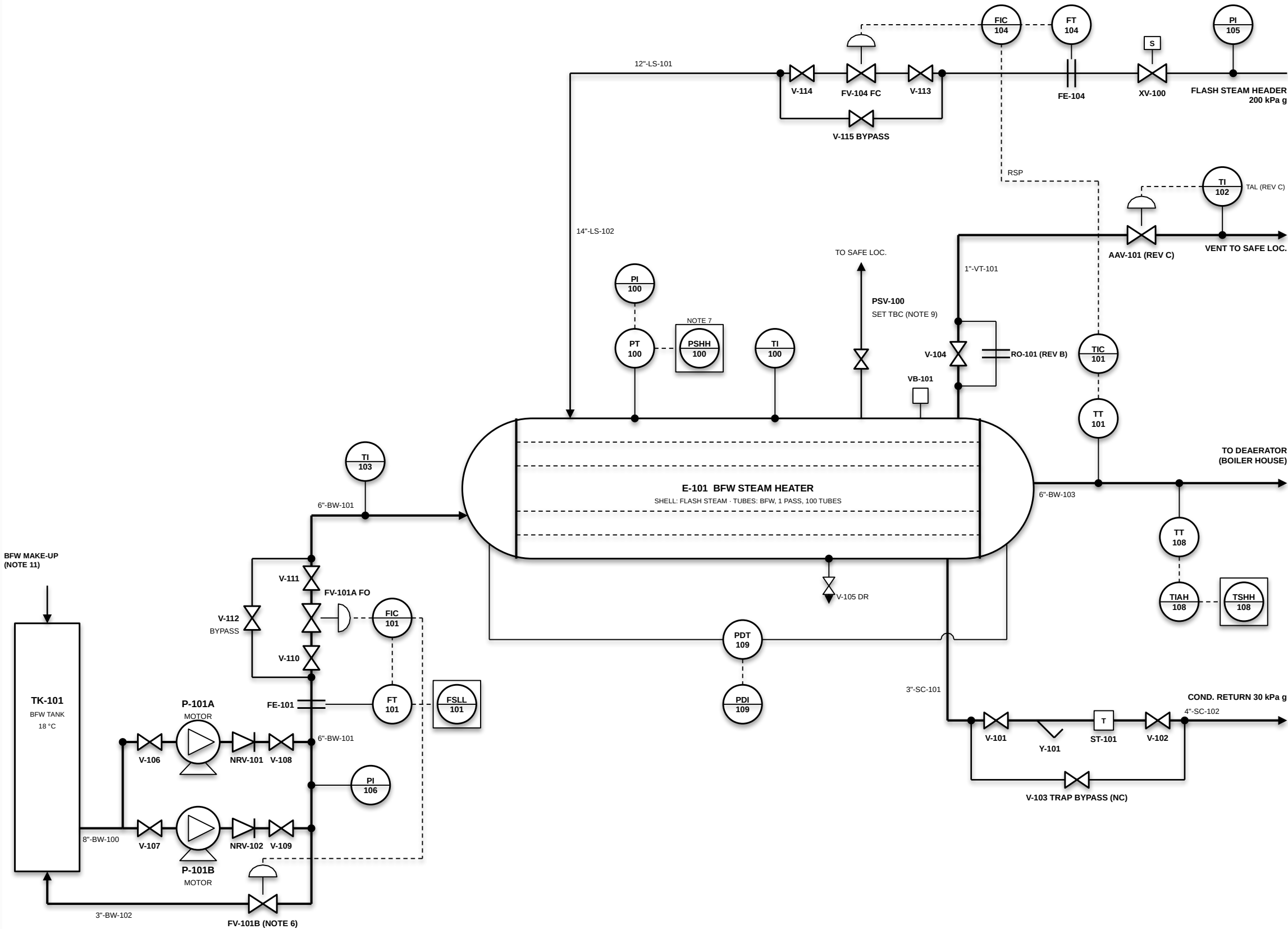




GENERAL NOTES:

1. DESIGN BASIS: BFW STEAM HEATER VER 1 OTS MODEL, DESIGN CASE. PRESSURES kPa g.
BFW 100 m³/h (27.7 kg/s) 18 → 70 °C; DUTY 6.04 MW. FLASH STEAM 9 935 kg/h FROM THE 200 kPa g HEADER (PI-105 195), SHELL 105 kPa g / 121.3 °C (λ 2 189 kJ/kg). P-101 DISCHARGE PI-106 662; TUBE-SIDE ΔP PDI-109 60 kPa. CONDENSATE TO THE 30 kPa g RETURN; HOTWELL LEVEL ≈ 5 % OF SHELL DIAMETER (TUBES START AT 12 %).

2. LINE SIZES FROM VELOCITY (BFW 6 in 1.5 m/s; STEAM 12 in 24 m/s, 14 in 28 m/s; CONDENSATE 3 in 0.6 m/s) - TBC.

3. PRESSURE INSTRUMENTS C/W ROOT AND TEST VALVES; PUMP CASING AND CHANNEL VENTS/DRAINS; TYP., NOT SHOWN.

4. XV-100 FAILS CLOSED, OPERATED BY SIS I-100 (SIGNALS NOT SHOWN). FV-104 FC, FV-101A FO.

5. TIC-101 SETS THE FIC-104 STEAM-FLOW SET POINT (CASCADE, RSP).

6. FV-101B MINIMUM-FLOW RECYCLE OPENS AS 100 – FIC-101 OUT (SPLIT); FAIL POSITION NOT IN THE OTS MODEL - TBC.

7. PSHH-100 SHARES PT-100 WITH PI-100, FSHL-101 SHARES FT-101 WITH FIC-101: OWN TRANSMITTERS RECOMMENDED - HOLD. TSHH-108 / TIAH-108 HAVE THEIR OWN TT-108, INDEPENDENT OF TT-101.

NOTES (CONT.):

8. NON-CONDENSABLES COLLECT AT THE SHELL END FURTHEST FROM THE STEAM INLET; V-104 VENTS THERE.
REV B: RO-101 CONTINUOUS VENT. REV C: AAV-101 AUTOMATIC AIR VENT WITH TI-102 / TAL-102.

9. PSV-100 SET PRESSURE AND SHELL / TUBE DESIGN PRESSURES ARE NOT IN THE OTS DATA (350 / 1 100 kPa g ASSUMED IN THE DESIGN TAB) - TBC. TUBE RUPTURE IS A RELIEF CASE BEYOND PSV-100: RUPTURE DISC RD-100 RECOMMENDED, HOLD.

10. VB-101 VACUUM BREAKER ADMITS AIR WHEN THE SHELL GOES BELOW ATMOSPHERIC (CONDENSATE HOLD-UP, WATER HAMMER).

11. TK-101 MAKE-UP AND LEVEL CONTROL OUTSIDE THE OTS SCOPE. HOTWELL LEVEL INDICATION NOT FITTED - RECOMMENDED, HOLD.

12. V-103 TRAP BYPASS NORMALLY CLOSED; OPEN IT PASSES LIVE STEAM TO THE RETURN HEADER.

13. ALARM SET POINTS: SEE WORKBOOK (ALARM TABLE); TRIP SET POINTS: INTERLOCK SCHEDULE.

14. V-106 TO V-115 AND NRV-101/102 ARE NUMBERED ON THIS SHEET; V-101 TO V-105 AS ON THE SCREEN DRAWING.

LEGEND

PROCESS LINE

UTILITY LINE (STEAM, VENT, RELIEF)

INSTRUMENT SIGNAL

MANUAL VALVE (GATE)

CONTROL VALVE, FO / FC = FAIL OPEN / CLOSED

SIS SHUT-OFF VALVE, SOLENOID, FAIL CLOSED

CHECK VALVE

FLOW ELEMENT (ORIFICE)

STEAM TRAP

RELIEF VALVE (PSV)

FIELD-MOUNTED INSTRUMENT

DCS FUNCTION (OPERATOR CONSOLE)

SIS FUNCTION

SERVICE CODES: BW BOILER FEED WATER · LS FLASH STEAM
SC STEAM CONDENSATE · VT VENT · DR DRAIN
LINE NO. = SIZE - SERVICE - NUMBER (SIZES TBC, NOTE 2)

INTERLOCK SCHEDULE · SIS I-100

TAG	CAUSE	SETTING	DELAY
FSHL-101	LOW BFW FLOW *	< 30 m³/h	5 s
TSHH-108	HIGH BFW OUTLET TEMP.	> 95 °C	5 s
PSHH-100	HIGH SHELL PRESSURE	> 250 kPa g	3 s
HS-ESD	ESD PUSHBUTTON (DCS)	MANUAL	0 s
PSV-100	E-101 SHELL OVERPRESSURE	TBC	MECH.

EFFECTS (I-100):
ANY TRIP: CLOSE XV-100 (STEAM SHUT-OFF);
FIC-104 TO MAN 0 %; TIC-101 TO MAN.
* ARMED ONLY WHEN THE SIS IS IN RUN (STEAM ON).
RESET BLOCKED WHILE TSHH-108 OR PSHH-100 IS ACTIVE.
OPEN XV-100 NEEDS: SIS RESET (IDLE), FIC-101 ≥ 50 m³/h,
FIC-104 IN MAN ≤ 10 %, V-101 AND V-102 OPEN.

REV	DESCRIPTION	DATE
B	RO-101 CONTINUOUS VENT (OPTION)	--
C	AAV-101 + TI-102; OTS SCREEN DWG (USED IN DECKS)	--
D	ALTERNATIVE VERSION, EPC STANDARD	03-OCT-26

BFW STEAM HEATER VER 1 · OTS			
P&ID — E-101 · ALTERNATIVE VERSION			
DRAWN: APINAN SOOTTITANTAWAT	CHK'D: --	APP'D: --	DATE: 03-OCT-26
DWG NO. BFWH-V1-PID-001	REV D	SCALE N.T.S.	SHEET 1 OF 1